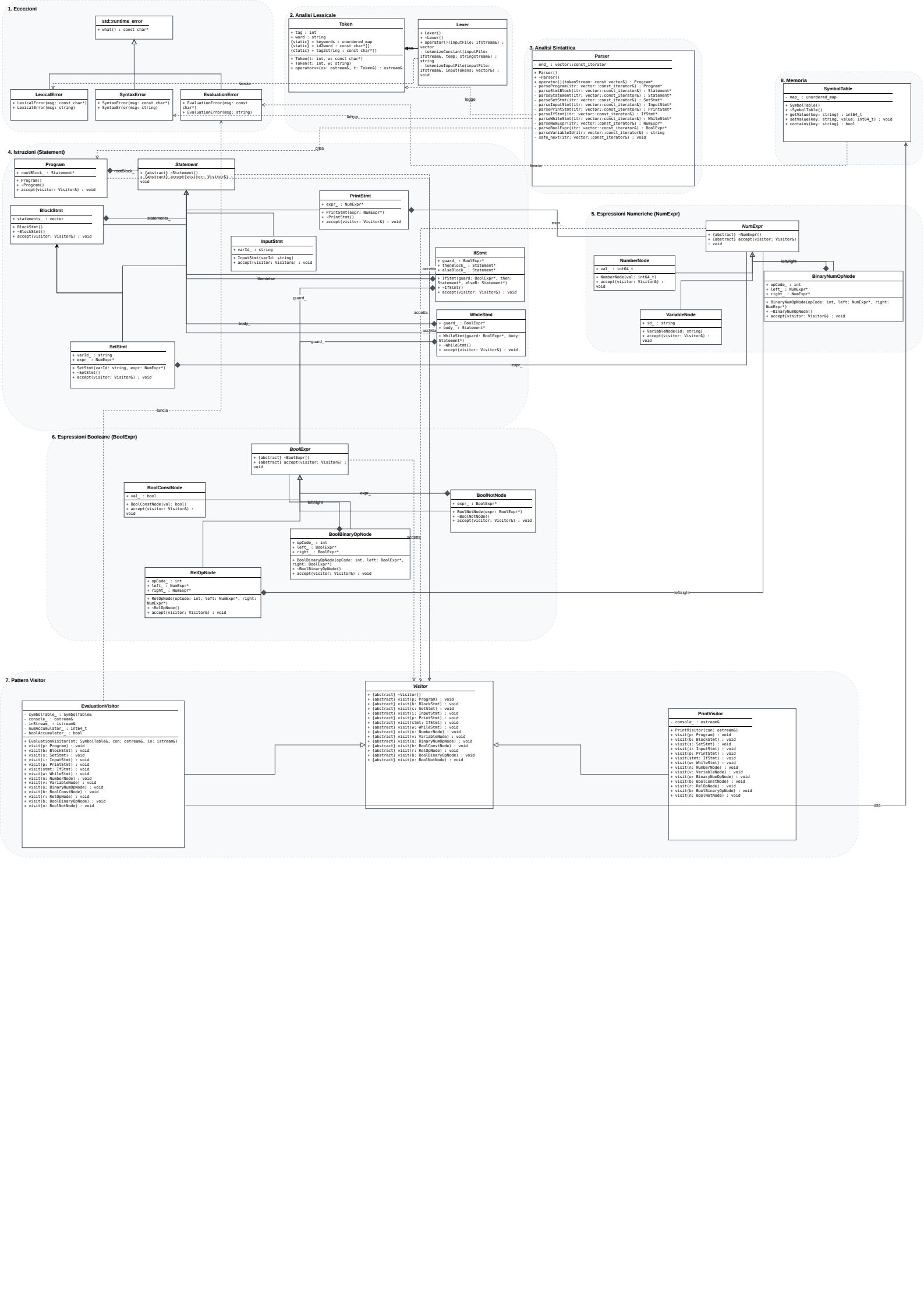

1. Eccezioni
std::runtime_error
+ what() : const char*
LexicalError
+ LexicalError(msg: const char*)
+ LexicalError(msg: string)
SyntaxError
+ SyntaxError(msg: const char*)
+ SyntaxError(msg: string)
EvaluationError
+ EvaluationError(msg: const char*)
+ EvaluationError(msg: string)
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2. Analisi Lessicale
Token
+ tag : int
+ word : string
{static} + keywords : unordered_map
{static} + idToword : const char*[]
{static} + tag2string : const char*[]
+ Token(t: int, w: const char*)
+ Token(t: int, w: string)
+ operator<<(os: ostream&, t: Token&) : ostream&
Lexer
+ Lexer()
+ ~Lexer()
+ operator()(inputFile: ifstream&) : vector
- tokenizeConstant(inputFile: ifstream&, temp: stringstream&) : string
- tokenizeInputFile(inputFile: ifstream&, inputTokens: vector&) : void
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3. Analisi Sintattica
Parser
- end_ : vector::const_iterator
+ Parser()
+ ~Parser()
+ operator()(tokenStream: const vector&) : Program*
- parseProgram(itr: vector::const_iterator&) : Program*
- parseStmtBlock(itr: vector::const_iterator&) : Statement*
- parseStatement(itr: vector::const_iterator&) : Statement*
- parseSetStmt(itr: vector::const_iterator&) : SetStmt*
- parseInputStmt(itr: vector::const_iterator&) : InputStmt*
- parsePrintStmt(itr: vector::const_iterator&) : PrintStmt*
- parseIfStmt(itr: vector::const_iterator&) : IfStmt*
- parseWhileStmt(itr: vector::const_iterator&) : WhileStmt*
- parseNumExpr(itr: vector::const_iterator&) : NumExpr*
- parseBoolExpr(itr: vector::const_iterator&) : BoolExpr*
- parseVariableId(itr: vector::const_iterator&) : string
- safe_next(itr: vector::const_iterator&) : void
8. Memoria
SymbolTable
- map_ : unordered_map
+ SymbolTable()
+ ~SymbolTable()
+ getValue(key: string) : int64_t
+ setValue(key: string, value: int64_t) : void
+ contains(key: string) : bool
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4. Istruzioni (Statement)
Program
+ rootBlock_ : Statement*
+ Program()
+ ~Program()
+ accept(visitor: Visitor&) : void
rootBlock_
Statement
+ {abstract} ~Statement()
+ {abstract} accept(visitor: Visitor&) : void
BlockStmt
+ statements_ : vector
+ BlockStmt()
+ ~BlockStmt()
+ accept(visitor: Visitor&) : void
statements_
PrintStmt
+ expr_ : NumExpr*
+ PrintStmt(expr: NumExpr*)
+ ~PrintStmt()
+ accept(visitor: Visitor&) : void
expr_
InputStmt
+ varId_ : string
+ InputStmt(varId: string)
+ accept(visitor: Visitor&) : void
IfStmt
+ guard_ : BoolExpr*
+ thenBlock_ : Statement*
+ elseBlock_ : Statement*
+ IfStmt(guard: BoolExpr*, then: Statement*, else: Statement*)
+ ~IfStmt()
+ accept(visitor: Visitor&) : void
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then/else
guard_
WhileStmt
+ guard_ : BoolExpr*
+ body_ : Statement*
+ WhileStmt(guard: BoolExpr*, body: Statement*)
+ ~WhileStmt()
+ accept(visitor: Visitor&) : void
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body_
guard_
SetStmt
+ varId_ : string
+ expr_ : NumExpr*
+ SetStmt(varId: string, expr: NumExpr*)
+ ~SetStmt()
+ accept(visitor: Visitor&) : void
expr_
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5. Espressioni Numeriche (NumExpr)
NumExpr
+ {abstract} ~NumExpr()
+ {abstract} accept(visitor: Visitor&) : void
NumberNode
+ val_ : int64_t
+ NumberNode(val: int64_t)
+ accept(visitor: Visitor&) : void
BinaryNumOpNode
+ opCode_ : int
+ left_ : NumExpr*
+ right_ : NumExpr*
+ BinaryNumOpNode(opCode: int, left: NumExpr*, right: NumExpr*)
+ ~BinaryNumOpNode()
+ accept(visitor: Visitor&) : void
left/right
VariableNode
+ id_ : string
+ VariableNode(id: string)
+ accept(visitor: Visitor&) : void
6. Espressioni Booleane (BoolExpr)
BoolExpr
+ {abstract} ~BoolExpr()
+ {abstract} accept(visitor: Visitor&) : void
BoolConstNode
+ val_ : bool
+ BoolConstNode(val: bool)
+ accept(visitor: Visitor&) : void
BoolNotNode
+ expr_ : BoolExpr*
+ ~BoolNotNode()
+ accept(visitor: Visitor&) : void
expr_
left/right
BoolBinaryOpNode
+ opCode_ : int
+ left_ : BoolExpr*
+ right_ : BoolExpr*
+ BoolBinaryOpNode(opCode: int, left: BoolExpr*, right: BoolExpr*)
+ ~BoolBinaryOpNode()
+ accept(visitor: Visitor&) : void
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RelOpNode
+ opCode_ : int
+ left_ : NumExpr*
+ right_ : NumExpr*
+ RelOpNode(opCode: int, left: NumExpr*, right: NumExpr*)
+ ~RelOpNode()
+ accept(visitor: Visitor&) : void
left/right
7. Pattern Visitor
EvaluationVisitor
- symbolTable_ : SymbolTable&
- console_ : ostream&
- inStream_ : istream&
- numAccumulator_ : int64_t
- boolAccumulator_ : bool
+ EvaluationVisitor(st: SymbolTable&, con: ostream&, in: istream&)
+ visit(p: Program) : void
+ visit(b: BlockStmt) : void
+ visit(s: SetStmt) : void
+ visit(i: InputStmt) : void
+ visit(p: PrintStmt) : void
+ visit(stmt: IfStmt) : void
+ visit(w: WhileStmt) : void
+ visit(n: NumberNode) : void
+ visit(v: VariableNode) : void
+ visit(o: BinaryNumOpNode) : void
+ visit(b: BoolConstNode) : void
+ visit(r: RelOpNode) : void
+ visit(b: BoolBinaryOpNode) : void
+ visit(n: BoolNotNode) : void
Visitor
+ {abstract} ~Visitor()
+ {abstract} visit(p: Program) : void
+ {abstract} visit(b: BlockStmt) : void
+ {abstract} visit(s: SetStmt) : void
+ {abstract} visit(i: InputStmt) : void
+ {abstract} visit(p: PrintStmt) : void
+ {abstract} visit(stmt: IfStmt) : void
+ {abstract} visit(w: WhileStmt) : void
+ {abstract} visit(n: NumberNode) : void
+ {abstract} visit(v: VariableNode) : void
+ {abstract} visit(o: BinaryNumOpNode) : void
+ {abstract} visit(b: BoolConstNode) : void
+ {abstract} visit(r: RelOpNode) : void
+ {abstract} visit(b: BoolBinaryOpNode) : void
+ {abstract} visit(n: BoolNotNode) : void
PrintVisitor
- console_ : ostream&
+ PrintVisitor(con: ostream&)
+ visit(p: Program) : void
+ visit(b: BlockStmt) : void
+ visit(s: SetStmt) : void
+ visit(i: InputStmt) : void
+ visit(p: PrintStmt) : void
+ visit(stmt: IfStmt) : void
+ visit(w: WhileStmt) : void
+ visit(n: NumberNode) : void
+ visit(v: VariableNode) : void
+ visit(o: BinaryNumOpNode) : void
+ visit(b: BoolConstNode) : void
+ visit(r: RelOpNode) : void
+ visit(b: BoolBinaryOpNode) : void
+ visit(n: BoolNotNode) : void
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